TASK 04 – WRITE CORRECT EXPRESSIONS

Assignment: Design, Implement and Validate a Payment Calculator for a Village Milk Co-operative
Bloom's Level: K4 – Analyse
Marks: 4

1. Aim

To write correct Python expressions for calculating fat steps, SNF steps, milk rate per litre and the total amount payable. To understand operator precedence, rounding methods, comparison chaining and unexpected expression results.

2. Expressions Used in the Milk Payment Calculator

The base rate is ₹30.00 per litre.

- Base fat = 3.5%
- Base SNF = 8.5%
- Fat step value = ₹0.40 for every 0.1%
- SNF step value = ₹0.30 for every 0.1%

The expressions are:

fat_steps = round((fat - 3.5) / 0.1)

snf_steps = round((snf - 8.5) / 0.1)

rate = 30.00 + (fat_steps * 0.40) + (snf_steps * 0.30)

amount = round(litres * rate, 2)

The round() function is used to get the nearest whole number of 0.1% steps.

3. Operator Precedence and Associativity

Operator precedence decides which operation is performed first in an expression.

Example:

result = 10 + 2 * 3

Output: 16

Multiplication is performed before addition.

Parentheses can change the order:

result = (10 + 2) * 3

Output: 36

Exponentiation uses **. It has higher precedence than unary minus.

```
print(-2 ** 2)

print((-2) ** 2)
```

Output:

-4

4

The first expression is evaluated as -(2 ** 2). The second squares the negative number.

4. Floating-Point Step Calculation

The expression:

```
print((4.2 - 3.5) / 0.1)
```

may produce:

7.000000000000001

This happens because decimal values such as 4.2 and 0.1 cannot always be represented exactly in binary floating-point format.

We use round() to correct the step count:

```
fat_steps = round((4.2 - 3.5) / 0.1)

print(fat_steps)
```

Output:

7

5. Difference Between round(), int() and math.floor()

- round() returns the nearest integer.
- int() removes the decimal part and moves towards zero.
- math.floor() returns the greatest integer less than or equal to the value.

Example:

```
import math


print(round(-3.8))

print(int(-3.8))

print(math.floor(-3.8))
```

Output:

-4

-3

-4

For negative numbers, int() and math.floor() give different results.

6. Comparison Chaining and Boolean Operators

Python allows multiple comparisons in one expression.

Example:

fat = 4.2

print(2.0 <= fat <= 7.0)

Output:

True

This checks whether fat is between 2.0 and 7.0, including both limits.

Boolean operators:

- and – True when both conditions are True.
- or – True when at least one condition is True.
- not – Reverses the Boolean result.

Example:

fat = 4.2

snf = 8.7

print(2.0 <= fat <= 7.0 and 6.0 <= snf <= 10.0)

Output:

True

7. Three Unexpected Expression Results

Example 1: Floating-point calculation

print(0.1 + 0.2)

Output may be:

0.30000000000000004

Reason: Some decimal numbers cannot be stored exactly in binary floating-point format.

Example 2: Negative integer division

print(-7 // 2)

Output:

-4

Reason: // performs floor division, so the result is rounded down towards negative infinity.

Example 3: Exponentiation and unary minus

print(-2 ** 2)

Output:

-4

Reason: Exponentiation is performed before unary minus.

8. Alternative Concept and Complexity Analysis

The number of fat steps can be calculated using round(), int() or math.floor().

For this comparison, SNF is kept at its base value of 8.5%, so only the fat adjustment changes the base rate of ₹30.00.

Fat	Method	Steps	Rate per litre
4.2	round()	7	₹32.80
4.2	int()	7	₹32.80
4.2	math.floor()	7	₹32.80
3.1	round()	-4	₹28.40
3.1	int()	-3	₹28.80
3.1	math.floor()	-4	₹28.40

For fat = 4.2, all three methods give the same result.

For fat = 3.1, int() gives -3 because it truncates towards zero. round() and math.floor() give -4.

Complexity:

- Time complexity: O(1), because each method performs a fixed number of operations.
- Space complexity: O(1), because only a fixed number of variables are used.

Therefore, round() is suitable for counting the nearest 0.1% step in this calculator.

9. Result

Correct Python expressions were identified for calculating fat steps, SNF steps, milk rate and payment amount. Operator precedence, floating-point errors, comparison chaining and the differences between round(), int() and math.floor() were demonstrated.